

Professional summary

Data Scientist and emerging AI Engineer with 5+ years of experience across aviation and energy, delivering production-grade data products, machine learning models, APIs, and impactful visualizations. Adept at leveraging SQL, Python, Power BI, LLMs, RAG, LangChain, and agentic AI workflows to optimize data infrastructure, enhance analytics, and streamline decision-making processes.

Skills

- **Programming:** - Python, R, SQL, VBA, C, Java, JavaScript
 - **Databases:** - Advanced SQL, PostgreSQL, BigQuery, MongoDB, Redis, Cassandra, NoSQL
 - **Machine Learning Tools (ML):** - Scikit-Learn, PyTorch, TensorFlow, forecasting, CNNs, RNNs, ARIMA
 - **Cloud & Deployment Tools:** - Microsoft Azure, AWS analytics, FastAPI, Flask
 - **AI Engineering:** - LLMs, RAG, LangChain, agentic AI, prompt engineering, embeddings, vector search
 - **IDE:** - Visual Studio Code, Antigravity, Cursor
- **Visualization Tools:** - Tableau, Power BI, R Shiny
 - **Web Development Tools:** - HTML5, CSS, JavaScript
 - **Version Control:** - Bitbucket
 - **Microsoft Office Suite:** - Word, PowerPoint, Excel
 - **Advanced Excel:** - Macros, pivot tables, VLOOKUP, XLOOKUP
 - **AI Tools:** - Claude, Gemini, AI-assisted automation

Work history

Senior Data Scientist IBA group, Aviation Sector • Spearheaded building forecasting model using best-fit machine learning programs. • Managing database integrity of various in-house Pythonic models. • Applying LLM and agentic AI workflows to support research, documentation, quality checks, and productivity.	01/2026 - Current
Data Scientist IBA group, Aviation Sector • Designed and implemented scalable data engineering infrastructure, optimising data storage, retrieval, and performance. • Developed production-grade Python APIs for data manipulation, modelling, and deployment-ready analytics solutions. • Delivered a robust quarterly update tool enabling consultants to efficiently update databases and generate aircraft value opinions. • Built and applied continuous testing of BigQuery datasets and tables, ensuring data integrity and accessibility for stakeholders. • Led data preparation and statistical modelling initiatives to support machine learning projects and advanced analytics use cases. • Migrated complex Excel-based models to Python, significantly improving scalability and performance on large datasets. • Deployed Power BI dashboards connected to AWS data sources, improving reporting efficiency. • Established end-to-end ETL pipelines and data warehousing solutions, reducing data delivery timelines. • Translated complex analytical findings into clear, actionable insights for senior leadership and clients. • Designed and maintained technical dashboards in Power BI, ensuring real-time status tracking. • Conducted validation and data integrity checks, ensuring high-quality analytics reports. • Incorporated agentic AI techniques to enhance workplace productivity and repeatable analysis workflows. • Developed complex financial aircraft leasing model for lessors to assess exposure when leasing.	09/2022 - 01/2026
Data Analyst IOCL, Energy Sector • Created a centralized repository by working on data warehousing activity across ERP systems, CRM systems and spreadsheets, increasing downstream productivity. • Developed a forecasting model using TensorFlow, reducing asset replacement downtime. • Built refinery data-driven Power BI dashboards, recognized by management and earning Employee of the Month twice consecutively. • Designed a memory-optimized database, reducing storage costs by 30%. • Operated in an Agile environment, participating in sprint planning, daily stand-ups, and cross-functional collaborations.	10/2020 - 06/2021
Graduate Apprentice Engineer IOCL, Energy Sector • Automated workflows using VBA and Macros, enhancing operational efficiency. • Built Excel loaders and Python scripts to interact with the database, keeping it up to date with the latest data. • Developed an Excel-based financial modelling tool for clients, later offered on a subscription basis. • Collaborated with product design teams to develop data-driven insights that enhanced user experience and business performance.	11/2019 - 10/2020
Education	
Master of Science: Business Analytics University Of Exeter - Exeter, UK • Awarded Distinction	05/2022
Bachelor of Technology Indian Institute Of Technology, BHU - Varanasi, India • GPA: 8.55	05/2017
A Level Equivalent Delhi Public School - Varanasi, India • GPA: 93.60	05/2012
GCSE Equivalent Delhi Public School - Varanasi, India • GPA: 93.1	05/2010

Projects

LLM-powered Aviation Knowledge Assistant (2026)

- Designed a RAG-style assistant concept using document ingestion, embeddings, vector search, prompt templates, and source-grounded responses.
- Focused on reducing manual lookup time for aviation valuation and analytics teams.

Agentic Analytics Workflow Automation (2026)

- Prototyped agentic AI workflows for recurring analysis tasks using planning prompts, Python execution, validation checks, and report-ready outputs.

Database Development Project (Jan 2022 - Apr 2022)

- Implemented polyglot persistence for an e-learning platform using SQLite and Redis.
- Applied privacy measures to safeguard personal identity data.

Machine Learning Application Project (Sep 2021 - Apr 2022)

- Developed an image classification model using CNNs and TensorFlow for an environmental analytics project.
- Performed descriptive, inferential, and predictive analysis on student performance datasets.

Interactive Financial Dashboard (Jan 2022 - Apr 2022)

- Designed an interactive financial dashboard using R Shiny, hosted on an R Shiny server.

Master's Dissertation: Using Song sentiment from Spotify to draw financial insights (2022)

- Retrieved song sentiment data via Spotify API and performed preprocessing including null imputation, aggregation, and filtering.
- Applied machine learning models including ARIMA and RNNs to forecast financial trends using Spotify metrics.